

Cloud Front- Route53-Task

Assignment Overview

This document covers seven independent tasks: peering two VPCs across regions, purchasing a domain from GoDaddy, deploying a static website in S3, creating a CloudFront CDN with an SSL certificate, creating a Route 53 hosted zone and mapping the domain to the CDN, updating the site content and confirming it through the domain, and sharing the domain in Slack for team verification. Each task resets and is documented with its own independent steps.

The following resource reference is used throughout this document:

Task	Resource	Purpose
1	VPC-A (ap-south-1) ↔ VPC-B (us-east-1)	Cross-region VPC peering
2	mycompanylab.com (example)	Domain purchased via GoDaddy
3	mydomain-site-bucket-2026	S3 static website
4	CloudFront distribution + ACM certificate	CDN with HTTPS
5	Route 53 hosted zone for mycompanylab.com	DNS mapped to CloudFront
6	index.html in mydomain-site-bucket-2026	Updated content served via domain

Task 1: Configure VPC Peering Across Regions

Outcome

VPC-A in ap-south-1 (10.0.0.0/16) and VPC-B in us-east-1 (10.1.0.0/16) are connected through an active inter-region VPC peering connection, with both route tables updated so instances in either VPC can reach each other by private IP.

Objective

To allow private, non-internet-routed communication between resources in two VPCs located in different AWS regions.

Prerequisites

- Two existing VPCs with non-overlapping CIDR blocks, one per region.
- IAM permissions to create and accept VPC peering connections and edit route tables.

Step-by-Step Implementation

Step 1: Request the Peering Connection from VPC-A's Region

VPC Console (ap-south-1) → Peering connections → Create peering connection → configure:

Parameter	Value
Name	VPC-A-to-VPC-B-Peering
VPC (Requester)	VPC-A (192.168.0.0/24)
Account	My account
Region	Another region → US East (N. Virginia) us-east-1
VPC (Acceptor)	VPC-B ID (172.168.0.0/24)

Click “Create peering connection.”

Screenshot Evidence:

aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) JT (565398309825) JT

VPC > Your VPCs > vpc-04a2ff9d365e436d0

VPC dashboard <

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

vpc-04a2ff9d365e436d0 / vpc1-n.virgina

Actions

Details Info

VPC ID vpc-04a2ff9d365e436d0	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0c71fc5639e429bd1	Main route table rtb-0089563a07152347b
Main network ACL acl-0d63b8c91feb4b49d	Default VPC No	IPv4 CIDR 192.168.0.0/24	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 565398309825
Encryption control ID -	Encryption control mode -		

Resource map CIDRs Flow logs Tags Related resources Integrations

Resource map Info

Show all details

vpc-0e570f764db9c6ee1 / my-vpc-Ohio

Actions

Details Info

VPC ID vpc-0e570f764db9c6ee1	State Available	Block Public Access Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0786b15af9269b066	Main route table rtb-00e59179e088de2f9
Main network ACL acl-0adb6aa7a7db156a4	Default VPC No	IPv4 CIDR 172.168.0.0/24	IPv6 pool -
IPv6 CIDR -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 565398309825
Encryption control ID -	Encryption control mode -		

Resource map CIDRs Flow logs Tags Related resources Integrations

Resource map Info

Show all details

VPC Your AWS virtual network my-vpc-Ohio	Subnets (0) Subnets within this VPC	Route tables (1) Route network traffic to resources rtb-00e59179e088de2f9	Network Connections Connections to other networks
---	---	--	---

Step 2: Accept the Peering Connection in VPC-B's Region

Switch console region to us-east-1 → VPC → Peering connections → select the pending request → Actions → Accept request → confirm.

Screenshot Evidence:

VPC > Peering connections > Create peering connection

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Select a local VPC to peer with

VPC ID (Requester)

vpc-04a2ff9d365e436d0 (vpc1-n.virgina) ▾

VPC CIDRs for vpc-04a2ff9d365e436d0 (vpc1-n.virgina)

CIDR	Status	Status reason
192.168.0.0/24	✔ Associated	-

Step 3: Update VPC-A's Route Table

ap-south-1 → Route Tables → select VPC-A's route table → Routes → Edit routes → Add route: Destination 10.1.0.0/16, Target: Peering connection → select the peering connection → Save.

Screenshot Evidence:

VPC > Peering connections > Create peering connection

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Select a local VPC to peer with

VPC ID (Requester)

vpc-04a2ff9d365e436d0 (vpc1-n.virgina) ▾

VPC CIDRs for vpc-04a2ff9d365e436d0 (vpc1-n.virgina)

CIDR	Status	Status reason
192.168.0.0/24	✔ Associated	-

Peering connections (1/1) [Info](#)

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Accepter CIDRs
-	pcx-0f501e45e43f2b304	Pending acceptance	vpc-04a2ff9d365e436d0	vpc-0e570f764db9c6ee1 / my-vpc-Ohio	192.168.0.0/24	-

Accept VPC peering connection request [Info](#)

Are you sure you want to accept this VPC peering connection request? (pcx-0f501e45e43f2b304)

Requester VPC

vpc-04a2ff9d365e436d0

Requester CIDRs

192.168.0.0/24

Requester Region

N. Virginia (us-east-1)

Requester owner ID

vpc-04a2ff9d365e436d0 (This account)

Accepter VPC

vpc-0e570f764db9c6ee1 / my-vpc-Ohio

Accepter Region

Ohio (us-east-2)

Accepter owner ID

vpc-0e570f764db9c6ee1 (This account)

Cancel

Accept request

pcx-0f501e45e43f2b304

Details

DNS

Route tables

Tags

Details

Step 4: Update VPC-B's Route Table

us-east-1 → Route Tables → select VPC-B's route table → Routes → Edit routes → Add route: Destination 10.0.0.0/16, Target: Peering connection → select the same peering connection → Save.

Screenshot Evidence:

rtb-0240ac415c2d8349f / public-rt

Actions

Details

Route table ID
rtb-0240ac415c2d8349f

VPC
vpc-0e570f764db9c6ee1 | my-vpc-Ohio

Main

No

Owner ID
565398309825

Explicit subnet associations

subnet-0bf5503d33e544913 / public-subnet

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (3)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0ab37d38348011650	Active	No	Create Route
172.168.0.0/24	local	Active	No	Create Route Table
192.168.0.0/24	pcx-0f501e45e43f2b304	Active	No	Create Route

Step 5: Allow Traffic in Security Groups

On instances in both VPCs, add inbound security group rules permitting traffic from the peer VPC's CIDR (e.g., allow ICMP/SSH from 192.168.0.0/24 on VPC-A instances, and from 172.168.0.0/24 on VPC-B instances).

Validation Steps

- Peering Connections console shows status “Active” for VPC-A-to-VPC-B-Peering.
- An EC2 instance in VPC-A can ping or SSH to an instance's private IP in VPC-B, and vice versa.

Explanation

VPC peering establishes a direct, private network connection between two VPCs using AWS's internal backbone, so traffic never traverses the public internet even across regions. Both sides' route tables must explicitly point to the peering connection for the peer's CIDR, since peering alone does not automatically route traffic — this two-sided route configuration is what makes the connection usable rather than just “Active.”

Real-Time Use Case

Organizations with workloads split across regions — for example, an application tier in one region and a shared services or database tier in another for latency, redundancy, or compliance reasons — use inter-region peering so those tiers can communicate privately and securely.

Common Errors and Troubleshooting

Error	Cause	Resolution
Peering connection stuck “Pending acceptance”	The accepter side has not yet approved the request	Switch to the accepter region/account and accept the peering connection
Instances still cannot reach each other after peering is “Active”	One or both route tables are missing the route to the peer CIDR via the peering connection	Add the missing route in whichever VPC's route table was not updated

Error	Cause	Resolution
“CIDR blocks overlap” error when creating the peering connection	Both VPCs use the same or overlapping IP range	Recreate one VPC with a non-overlapping CIDR block (e.g., 10.0.0.0/16 and 10.1.0.0/16)

Task 2: Purchase One Domain from GoDaddy

Outcome

A domain (e.g., mycompanylab.com) is purchased and active under a GoDaddy account, visible in the GoDaddy “My Products” domain list with a valid expiration date.

Objective

To acquire a registered domain name that will later be pointed at the AWS-hosted website through Route 53.

Prerequisites

- A GoDaddy account (or account created during purchase).
- A valid payment method.
- A candidate domain name to search for.

Step-by-Step Implementation

Step 1: Search for the Domain

Go to godaddy.com → enter the desired name (e.g., mycompanylab.com) in the domain search bar → click “Search.”

Step 2: Select and Add the Domain to Cart

From the search results, choose the exact available domain and extension → click “Add to Cart.”

Step 3: Configure Purchase Options

In the cart, review/adjust:

Parameter	Value
Registration period	1 year (minimum)
Domain privacy protection	Disable (recommended, hides WHOIS contact details)
Auto-renew	Disable (recommended, prevents accidental expiration)

Step 4: Complete Checkout

Proceed to checkout → sign in or create a GoDaddy account → enter payment details → complete the purchase.

Screenshot Evidence:

Domain

Connect Domain

krishna10.in

☆

Your Domain is Hot

✕

krishna10.in is attracting significant interest from other GoDaddy customers! Consider enabling auto-renew to maintain ownership.

[Turn Auto-renew On](#) →

- Overview
- DNS
- Registration Settings
- Products
- Activity Log

Renew

Renew Now

Auto-renew

Off

Expires On

Jan 22, 2027

Renewal Cost ⓘ

₹899.00/yr

Quick Links

🔗

[Manage DNS](#)

➡

[Forward Domain](#)

✉

[Connect Email](#)

\$

[Sell This Domain](#)

Validation Steps

- GoDaddy account → My Products → Domains shows the purchased domain listed as active.
- A confirmation email from GoDaddy is received for the purchase.

Explanation

Purchasing a domain registers it with a domain registrar (GoDaddy), which reserves the name and by default manages its DNS. This domain is the human-readable name that will later be delegated to Route 53 (Task 5) so AWS can control where it points, while GoDaddy remains the registrar of record.

Real-Time Use Case

Businesses register a domain through a registrar such as GoDaddy and then delegate DNS management to a cloud provider's DNS service (Route 53) so the same domain can point to cloud-hosted infrastructure such as a CDN, load balancer, or static website.

Common Errors and Troubleshooting

Error	Cause	Resolution
Desired domain name is unavailable	The exact name is already registered by someone else	Try a different extension (.net, .io) or a variation of the name
Payment declined at checkout	Card details incorrect or blocked by the bank for an online purchase	Verify card details, or use an alternate payment method
Domain shows “Pending” instead of “Active” after purchase	Registration is still processing (usually completes within minutes)	Wait a few minutes and refresh the My Products page

Task 3: Deploy a Static Website in S3

Outcome

A bucket named mydomain-site-bucket-2026 is configured for static website hosting with an index.html uploaded, and the site is reachable at its S3 website endpoint URL.

Objective

To host the website's content on S3 so it can later be served securely and globally through a CDN mapped to the purchased domain.

Prerequisites

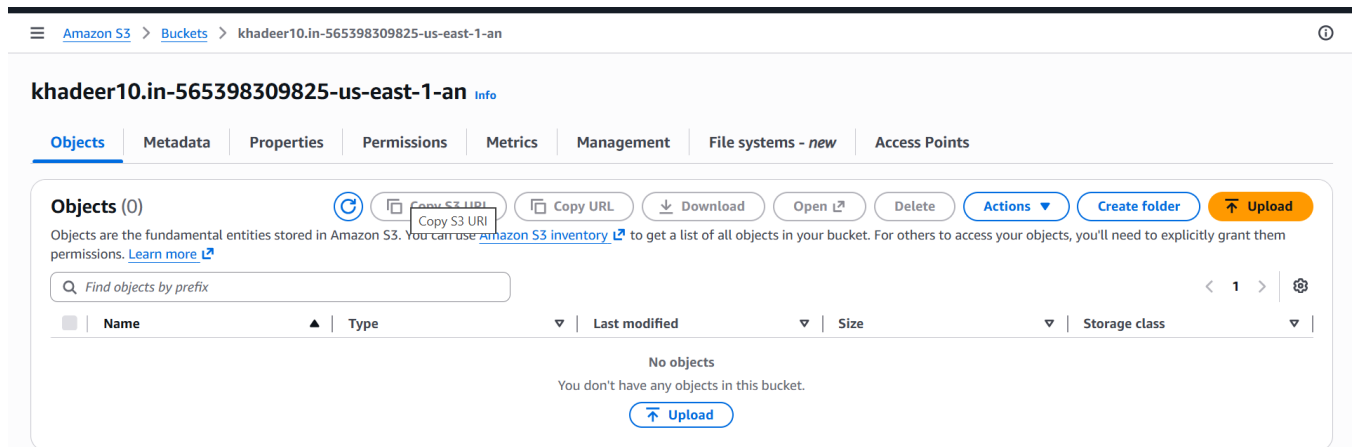
- AWS account with S3 permissions.
- An index.html file (and optionally error.html) prepared locally.

Step-by-Step Implementation

Step 1: Create the Bucket

S3 Console → Create bucket → Name: khadeer10.in → Region: us-east-1 → Create bucket.

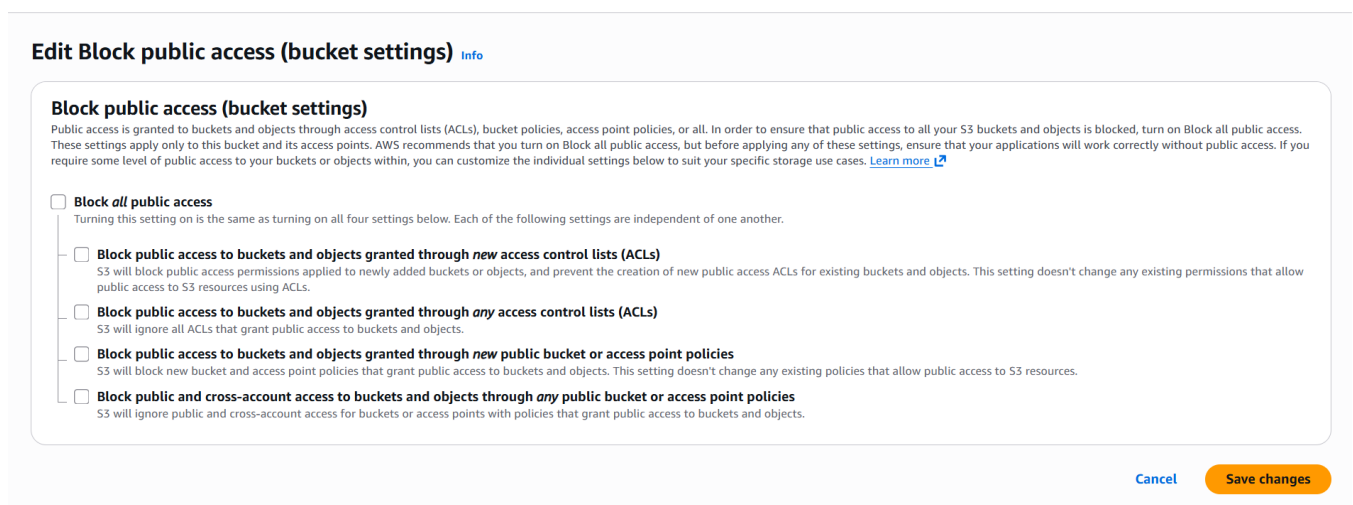
Screenshot Evidence:



Step 2: Unblock Public Access

Open the bucket → Permissions tab → Block public access → Edit → uncheck “Block all public access” → Save → confirm.

Screenshot Evidence:



Step 3: Upload the Website Files

Objects tab → Upload → Add files → select index.html (and error.html) → Upload.

Screenshot Evidence:

The screenshot shows the 'Upload: status' page in the Amazon S3 console. It displays a summary of the upload process, indicating that 2 files (23.1 KB) were successfully uploaded. Below the summary, there is a table titled 'Files and folders' showing the details of the uploaded files: error.html (2.9 KB) and index.html (20.2 KB). Both files are marked as 'Succeeded'.

Name	Folder	Type	Size	Status	Error
error.html	-	text/html	2.9 KB	Succeeded	-
index.html	-	text/html	20.2 KB	Succeeded	-

Step 4: Enable Static Website Hosting

Parameter	Value
Static website hosting	Enable
Index document	index.html
Error document	error.html

Properties tab → Static website hosting → Edit → apply the settings above → Save, and note the Bucket website endpoint.

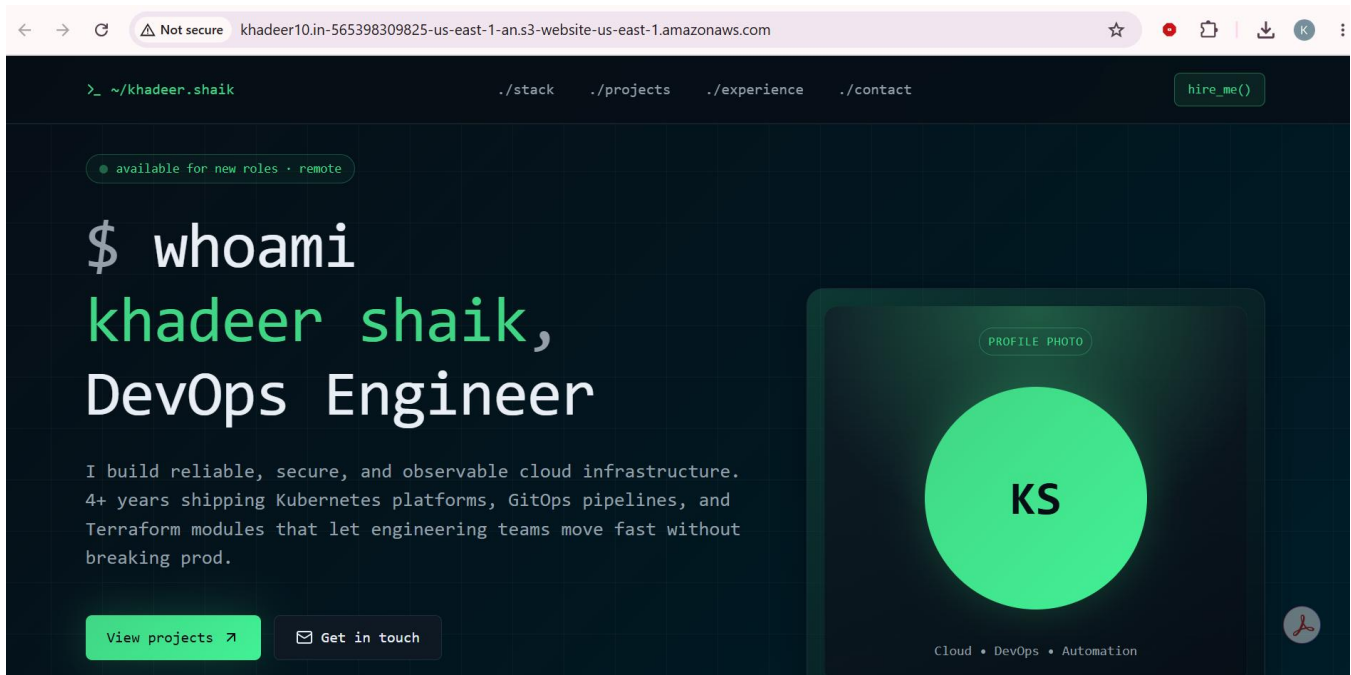
Screenshot Evidence:

The screenshot shows the 'Edit static website hosting' page in the Amazon S3 console. It displays the configuration for static website hosting, including the option to 'Enable' static website hosting and the 'Host a static website' option. The 'Index document' is set to 'index.html' and the 'Error document' is set to 'error.html'. A note indicates that for customers to access content at the website endpoint, the content must be made publicly readable.

Validation Steps

- Opening the S3 website endpoint URL in a browser displays the content of index.html.

Validation Screenshot:



Explanation

S3 static website hosting serves the site's files directly over HTTP without any web server or compute instance. This bucket becomes the origin that CloudFront (Task 4) will pull content from, so its endpoint and public-read policy must be working correctly before the CDN is layered on top.

Real-Time Use Case

Serving a static site from S3 behind a CDN is a common, low-cost architecture for marketing sites, documentation, and single-page applications that don't need server-side rendering.

Common Errors and Troubleshooting

Error	Cause	Resolution
"403 Forbidden" on the website endpoint	Block Public Access still enabled, or bucket policy not saved	Disable Block Public Access and confirm the public-read policy is saved
"NoSuchWebsiteConfiguration"	Static website hosting was never enabled	Properties tab → Static website hosting → Edit → Enable
Bucket name already taken	S3 bucket names are globally unique	Choose a different unique bucket name

Task 4: Create a CDN and Attach One SSL Certificate

Outcome

A CloudFront distribution is created with the S3 static website endpoint as its origin, an AWS Certificate Manager (ACM) SSL certificate for the domain is issued and attached, and the CloudFront domain name serves the site over HTTPS.

Objective

To distribute the website globally with low latency through a CDN, and to serve it securely over HTTPS using a valid SSL certificate.

Prerequisites

- Task 3 completed — S3 static website is live.
- Task 2 completed — domain purchased and available to validate a certificate against.

- IAM permissions for CloudFront and ACM.

Step-by-Step Implementation

Step 1: Request an SSL Certificate in ACM (us-east-1)

Certificate Manager Console (region must be us-east-1 for CloudFront) → Request certificate → Request a public certificate → Domain name: mycompanylab.com (and www.mycompanylab.com) → Validation method: DNS validation → Request.

Screenshot Evidence:

The screenshot shows the AWS Certificate Manager console in the 'us-east-1' region. The breadcrumb trail is: AWS Certificate Manager > Certificates > Request certificate > Request public certificate. The main heading is 'Request public certificate'.

Domain names
Provide one or more domain names for your certificate.
Fully qualified domain name:
[Add another name to this certificate](#)
You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.

Allow export [Info](#)
☒ **Disable export**
Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.
☐ **Enable export**
Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.

Validation method [Info](#)
Select a method for validating domain ownership.
☒ **DNS validation - recommended**

The left sidebar shows the 'AWS Certificate Manager (ACM)' menu with options: List certificates, Request certificate, Import certificate, AWS Private CA, and ACME (with sub-options: ACME certificates, Endpoints).

The main content area shows the 'Certificates (1)' list with the following table:

Certificate ID	Status	Common name	Type	In use	Renewal
3fc1e5b3-7ae8-42ba-b079-49a70583bcc7	Pending validation	krishna10.in	Amazon Issued	No	Ineligible

Step 2: Validate the Certificate

ACM shows a CNAME record to create → copy the Name/Value → add it in the domain's DNS (temporarily via GoDaddy DNS, or later via Route 53 once delegated in Task 5) → wait for ACM status to change to “Issued.”

Screenshot Evidence:

This screenshot shows the same AWS Certificate Manager console as the previous one, but the certificate status has changed to 'Issued'.

The left sidebar remains the same, showing the 'AWS Certificate Manager (ACM)' menu.

The main content area shows the 'Certificates (1)' list with the following table:

Certificate ID	Status	Common name	Type	In use	Renewal
3fc1e5b3-7ae8-42ba-b079-49a70583bcc7	Issued	krishna10.in	Amazon Issued	No	Ineligible

Step 3: Create the CloudFront Distribution

Parameter	Value
Origin domain	Khadeer10.in-565398309825-us-east-1-an.s3-wesite-us-east-1.amazonaws.com
Viewer protocol policy	Redirect HTTP to HTTPS
Alternate domain name (CNAME)	mycompanylab.com
Custom SSL certificate	Select the ACM certificate issued in Step 2

CloudFront Console → Create distribution → apply the settings above → Create distribution.

Screenshot Evidence:

CloudFront > Distributions > Create distribution

Step 1
Get started

Step 2
Specify origin

Step 3
Enable security

Step 4
Get TLS certificate

Step 5
Review and create

Get started

Connect your websites, apps, files, video streams, and other content to CloudFront. We optimize the performance, reliability, and security for your web traffic.

Distribution options

Distribution name

Name will be stored as a tag on the resource. You can change the name, or more tags, later.

khadeer10.in

Description - optional

Distribution type

☒ Single website or app

Choose if each website or application will have a unique configuration.

☐ Multi-tenant architecture - New

Choose when you have multiple domains that need to share configurations. This is a common architecture for SaaS providers.

Domain

Route 53 managed domain - optional

Enter a domain that's already registered with Route 53 in your AWS account. CloudFront will provision a TLS certificate for you. If you have a domain from a different DNS provider, skip this step and configure your domain later.

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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CloudFront > Distributions > Create distribution

Step 1
Get started

Step 2
Specify origin

Step 3
Enable security

Step 4
Review and create

Specify origin

Origin type

Your origin is where your content (such as a website or app) lives. CloudFront works with AWS-based origins and origins hosted on other cloud providers.

☒ Amazon S3

Deliver static assets like files and images, statically generated websites or single page applications (SPA).

☐ Elastic Load Balancer

Deliver applications hosted behind ELB such as dynamic websites, web services, and APIs.

☐ API Gateway

Deliver API endpoints for REST APIs hosted on API Gateway.

☐ Elemental MediaPackage

Deliver end-to-end live events or video on demand (VOD).

☐ VPC origin

Deliver applications and content hosted within private VPCs, such as EC2 instances and Application Load Balancers.

☐ Other

Refer to any AWS or non-AWS origin through its publicly resolvable URL.

Origin

S3 origin

Choose an AWS origin, or enter your origin's domain name. [Learn more](#)

khadeer10.in-565398309825-us-east-1-an.s3-website-us-east-1.amazonaws.com [Browse S3](#)

Origin path - optional

The directory path within your origin where your content is stored. [Learn more](#)

Step 4: Wait for Deployment

Wait for the distribution's status to change from “Deploying” to “Enabled” (typically several minutes), then note the CloudFront domain name (e.g., d123456abcdef.cloudfront.net).

Screenshot Evidence:

CloudFront > Distributions > E2VR75VOIGG1FJ

krishna10.in Standard View metrics

Details

Distribution domain name

d2jj6ffceiqp24.cloudfront.net

Billing

-

ARN

arn:aws:cloudfront::565398309825:distribution/E2VR75VOIGG1FJ

Last modified

July 24, 2026 at 11:32:18 AM UTC

General

Security

Origins

Behaviors

Error pages

Invalidations

Logging

Tags

Settings

Name

krishna10.in

Description

-

Price class

Use all edge locations (best performance)

Supported HTTP versions

HTTP/2, HTTP/1.1, HTTP/1.0

Alternate domain names

krishna10.in

Route domains to CloudFront

Custom SSL certificate

krishna10.in

Security policy

TLV1.2_2021

Standard logging

Off

Cookie logging

Off

Default root object

-

Cache tag header

-

Continuous deployment info

Create staging distribution

Validation Steps

- Opening <https://d2jj6ffceiqp24.cloudfront.net> in a browser shows the website over HTTPS with a valid certificate (padlock icon, no browser warning).
- ACM Console shows the certificate status as “Issued” and associated with the distribution.

Explanation

CloudFront caches the site's content at edge locations around the world, reducing latency for visitors far from us-east-1, while ACM issues a free, auto-renewing SSL certificate that CloudFront uses to terminate HTTPS at the edge. The certificate must be requested in us-east-1 specifically, since CloudFront only accepts ACM certificates from that region regardless of which region the distribution otherwise targets.

Real-Time Use Case

Nearly every production website uses a CDN with a CDN-attached SSL certificate to combine fast global content delivery with the HTTPS encryption that browsers and users expect by default.

Common Errors and Troubleshooting

Error	Cause	Resolution
Certificate stuck “Pending validation”	The DNS validation CNAME record was not created, or not yet propagated	Add the exact CNAME record ACM provided in the domain's DNS and wait for propagation
CloudFront distribution deployed but shows the S3 XML error page	Origin domain points to the S3 REST endpoint instead of the website endpoint	Use the S3 static website endpoint (...s3-website-region.amazonaws.com) as the origin, not the bucket's REST endpoint
Browser shows “NET::ERR_CERT_COMMON_NAME_INVALID”	The alternate domain name (CNAME) on the distribution doesn't match the certificate's domain	Ensure the exact same domain name is used in both the ACM certificate and the CloudFront alternate domain name field

Task 5: Create a Route 53 Hosted Zone and Map the Domain to the CDN

Outcome

A Route 53 public hosted zone for mycompanylab.com is created, GoDaddy's nameservers are updated to point to Route 53, and an alias A record maps the domain directly to the CloudFront distribution.

Objective

To delegate DNS management of the purchased domain to Route 53 and point it at the CloudFront CDN, so visitors can reach the site using the domain name instead of the CloudFront URL.

Prerequisites

- Task 2 completed — domain purchased on GoDaddy.
- Task 4 completed — CloudFront distribution enabled.
- Access to the GoDaddy account's DNS/nameserver settings.

Step-by-Step Implementation

Step 1: Create the Hosted Zone

Route 53 Console → Hosted zones → Create hosted zone → Domain name: mycompanylab.com → Type: Public hosted zone → Create hosted zone.

Screenshot Evidence:

The screenshot shows the AWS Route 53 console's 'Create hosted zone' page. The breadcrumb navigation at the top reads 'Route 53 > Hosted zones > Create hosted zone'. The main heading is 'Create hosted zone' with an 'Info' link. Below this is the 'Hosted zone configuration' section, which includes a description: 'A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.' The 'Domain name' field is labeled 'Info' and contains 'krishna10.in'. A note below it says 'This is the name of the domain that you want to route traffic for.' and lists valid characters. The 'Description - optional' field is labeled 'Info' and contains 'The hosted zone is used for...'. A note below it says 'This value lets you distinguish hosted zones that have the same name.' and 'The description can have up to 256 characters. 0/256'. The 'Type' section is labeled 'Info' and has two radio buttons: 'Public hosted zone' (selected) and 'Private hosted zone'. The 'Public hosted zone' option has a note: 'A public hosted zone determines how traffic is routed on the internet.' The 'Private hosted zone' option has a note: 'A private hosted zone determines how traffic is routed within an Amazon VPC.'

Step 2: Copy the Route 53 Nameservers

Open the new hosted zone → copy the four NS record values (e.g., ns-123.awsdns-45.com).

Screenshot Evidence:

Public

krishna10.in

Info

Delete zone

Test record

Configure query logging

Hosted zone details

Edit hosted zone

Records (4)

Accelerated recovery

DNSSEC signing

Hosted zone tags (0)

Records (4) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Type

Routing p...

Alias

< 1 >

⚙️

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalua...	Recor...
☐	krishna10.in	A	Simple	-	Yes	d2jj6fciqp24.cloudfront.net.	-	-	No	-
☐	krishna10.in	NS	Simple	-	No	ns-586.awsdns-09.net. ns-1553.awsdns-02.co.uk. ns-1410.awsdns-48.org. ns-112.awsdns-14.com.	172,800	-	-	-
☐	krishna10.in	SOA	Simple	-	No	ns-586.awsdns-09.net. awsd...	900	-	-	-

Parameter	Value
Record type	A – Routes traffic to an IPv4 address
Alias	Yes
Route traffic to	Alias to CloudFront distribution → select the distribution from Task 4

Click “Create records.”

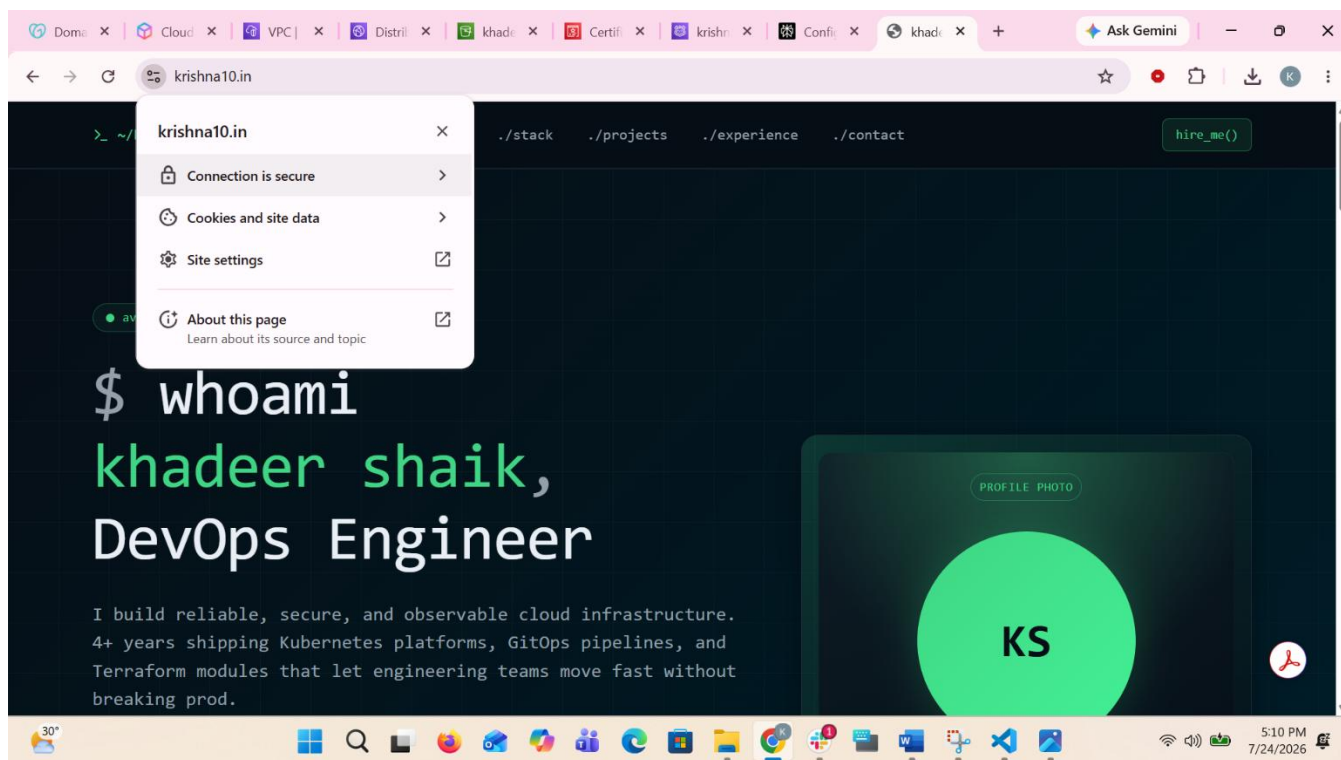
Screenshot Evidence:

The screenshot shows the AWS Route 53 console's 'Create record' page. The breadcrumb navigation at the top indicates the path: Route 53 > Hosted zones > krishna10.in > Create record. The main section is titled 'Create record' with an 'Info' link. Below this is a 'Quick create record' section. Under 'Record 1', there is a 'Record name' field with 'subdomain' and a 'Record type' dropdown set to 'A – Routes traffic to an IPv4 address and some AWS resources'. The 'Route traffic to' dropdown is set to 'Alias to CloudFront distribution', and the 'Routing policy' is set to 'Simple routing'. The 'Evaluate target health' toggle is turned off. A 'Delete' button is located next to the record name. At the bottom right, there is an 'Add another record' button.

Validation Steps

- A DNS lookup (e.g., dig mycompanylab.com or nslookup mycompanylab.com) resolves to a CloudFront-associated address, not GoDaddy's default parking page.
- Opening https://mycompanylab.com in a browser loads the same site served by the CloudFront distribution.

Validation Screenshot:



Explanation

Updating GoDaddy's nameservers to Route 53's delegates all DNS resolution for the domain to AWS, while GoDaddy remains only the registrar of record. The alias record is important because it lets Route 53 point the domain's root directly at a CloudFront distribution — something a traditional CNAME record cannot do at a zone apex — without needing a separate IP address to manage.

Real-Time Use Case

This registrar-plus-Route-53 pattern is the standard way organizations connect a domain bought from any registrar to AWS-hosted infrastructure, letting them keep their preferred registrar for billing/ownership while using Route 53 for all actual DNS routing.

Common Errors and Troubleshooting

Error	Cause	Resolution
Domain still resolves to GoDaddy's parking page	Nameserver change has not fully propagated yet, or was not saved correctly in GoDaddy	Wait up to 24–48 hours for propagation; re-check the nameserver values in GoDaddy
“Alias to CloudFront distribution” option doesn't show the distribution	The distribution is still “Deploying” and not yet “Enabled”	Wait for the CloudFront distribution status to become Enabled, then retry
Site loads over the CloudFront URL but not over the custom domain	The alias record was created in the wrong hosted zone, or points to a different distribution	Confirm the record is in the mycompanylab.com hosted zone and targets the correct distribution

Task 6: Update index.html in S3 and Verify via the Domain Name

Outcome

An updated index.html is uploaded to mydomain-site-bucket-2026, the CloudFront cache is invalidated, and the new content is visible when visiting <https://mycompanylab.com>.

Objective

To confirm the full content pipeline works end-to-end: an S3 update is reflected through CloudFront and reachable via the custom domain, not just the S3 or CloudFront URLs directly.

Prerequisites

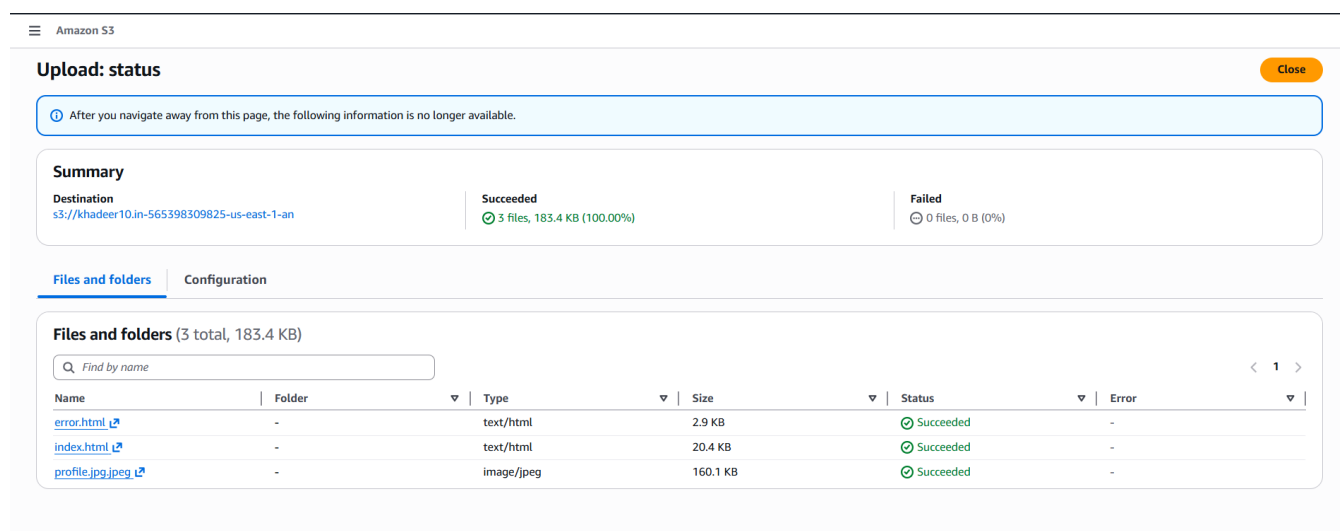
- Tasks 3–5 completed — site live, CDN enabled, domain mapped.
- An updated version of index.html ready to upload.

Step-by-Step Implementation

Step 1: Edit and Upload the Updated File

Modify index.html locally (e.g., change the heading text) → S3 Console → mydomain-site-bucket-2026 → Upload → select the updated index.html → Upload (overwriting the existing object).

Screenshot Evidence:



Amazon S3

Upload: status Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination
s3://khadeer10.in-565398309825-us-east-1-an

Succeeded
3 files, 183.4 KB (100.00%)

Failed
0 files, 0 B (0%)

Files and folders | Configuration

Files and folders (3 total, 183.4 KB)

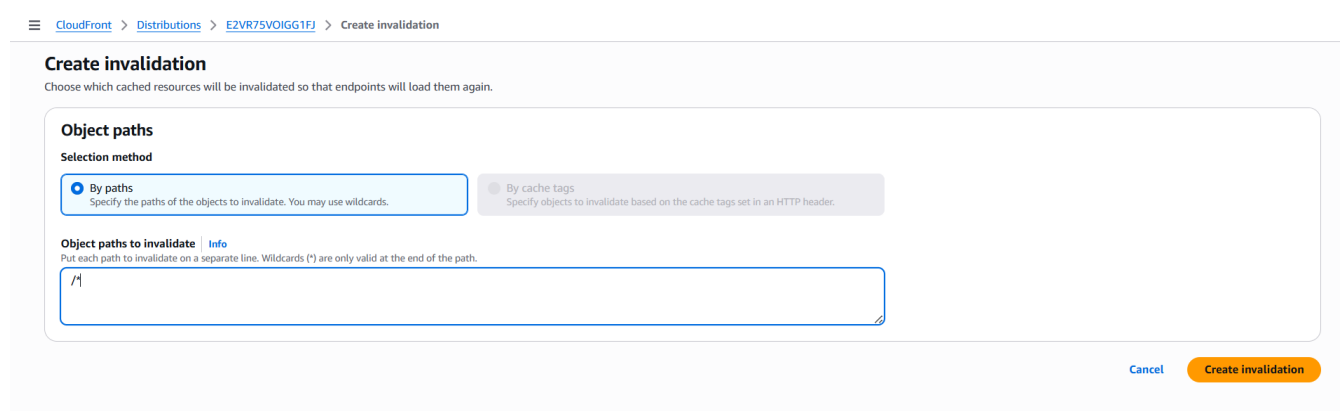
Find by name

Name	Folder	Type	Size	Status	Error
error.html	-	text/html	2.9 KB	Succeeded	-
index.html	-	text/html	20.4 KB	Succeeded	-
profile.jpg.jpeg	-	image/jpeg	160.1 KB	Succeeded	-

Step 2: Invalidate the CloudFront Cache

CloudFront Console → select the distribution → Invalidations tab → Create invalidation → Object path: /index.html (or /* for everything) → Create invalidation.

Screenshot Evidence:



CloudFront > Distributions > E2VR75VOIGG1FJ > Create invalidation

Create invalidation

Choose which cached resources will be invalidated so that endpoints will load them again.

Object paths

Selection method

☒ By paths
Specify the paths of the objects to invalidate. You may use wildcards.

☐ By cache tags
Specify objects to invalidate based on the cache tags set in an HTTP header.

Object paths to invalidate [Info](#)
Put each path to invalidate on a separate line. Wildcards (*) are only valid at the end of the path.

/*

Cancel Create invalidation

Step 3: Wait for the Invalidation to Complete

Wait until the invalidation's status changes to “Completed” (usually under a minute).

Screenshot Evidence:

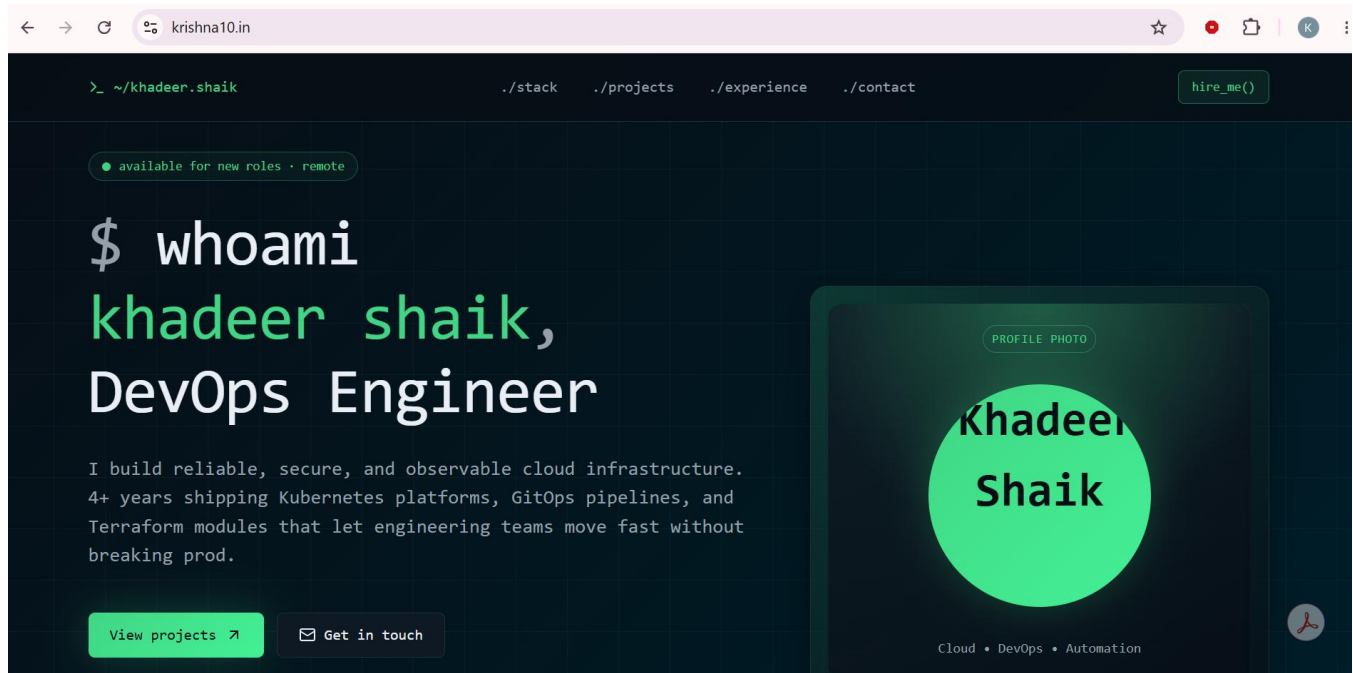


Step 4: Verify via the Domain Name

Open a browser (or an incognito window, to avoid local caching) and visit:

<https://krishna10.in>

Screenshot Evidence:



Validation Steps

- The page loads over <https://mycompanylab.com> and shows the newly updated content, not the previous version.

Explanation

Because CloudFront caches content at edge locations, simply overwriting the S3 object is not enough to make visitors see the change immediately — an invalidation forces CloudFront to fetch a fresh copy from the S3 origin on the next request. Verifying through the domain (rather than the CloudFront or S3 URLs directly) confirms every layer set up in Tasks 3 through 5 — S3, CloudFront, ACM, and Route 53 — is correctly connected end-to-end.

Real-Time Use Case

Content teams updating a CDN-fronted static site routinely follow this same upload-then-invalidate workflow (or automate it in a deployment pipeline) to ensure visitors see new content promptly instead of a stale cached version.

Common Errors and Troubleshooting

Error	Cause	Resolution
Domain still shows the old content after upload	CloudFront is still serving the cached version because no invalidation was created	Create a CloudFront invalidation for /index.html or /*
Change visible on the S3 endpoint but not on the domain	Invalidation not yet completed, or browser is showing a locally cached copy	Wait for the invalidation to complete and reload using a private/incognito window
Upload succeeds but page shows an error	New index.html has a syntax error or was uploaded with the wrong content type	Verify the HTML is valid and the object's Content-Type is text/html

Final Verification: Summary of All Seven Tasks

The table below summarizes the verification point for each task, confirming that the domain, CDN, and cross-region VPC setup were completed successfully:

Task	Action	Verification
1	Cross-region VPC peering	Instances in VPC-A and VPC-B reach each other by private IP
2	Purchase domain from GoDaddy	Domain listed as active in GoDaddy My Products
3	Deploy static website in S3	S3 website endpoint renders index.html
4	CDN with SSL certificate	CloudFront URL loads the site over HTTPS with a valid certificate
5	Route 53 hosted zone mapped to CDN	Domain resolves and loads the CloudFront-served site
6	Update index.html, verify via domain	https://domain shows the updated content after invalidation
7	Share domain in Slack	Teammates confirm the site loads from their own networks